Lab Report 6: The Quantum Eraser

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Experimental Setup Used: #1

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1 Abstract

In this experiment, we explore the phenomena of quantum interference and erase information using a quantum eraser setup based on a Mach-Zender Interferometer (MZI). Through manipulating polarization through half wave plates and controlling phase of single photons, we are able to investigate how entanglement between spatial and polarization degrees of freedom affects interference visibility. We found that interference disappears when path information is available and reappears when measured in the diagonal basis, effectively erasing the path label. These results highlight how choosing specific quantum measurements affect observable outcomes.

2 Introduction

Quantum mechanics challenges classical understandings of particle movement, especially when it comes to measurements. One of the important elements of quantum systems is that measurement can influence what is observed. When a particle in our case a single photon has more than one possible path to travel it can interfere with itself, producing an observable pattern that depends on the relative phases of those paths. This is known as quantum interference and can only occur when the paths are indistinguishable.

The quantum eraser experiment explores what happens when path information is available or by choice "erased". By entangling a single photon spatial path with polarization the paths become distinguishable and interference disappears. However, if the measurement is made on the basis that obscures path information, the interference pattern can be found again even after photon detection.

This experiment aims to demonstrate how to handle the path information whether we erase it or make it available through measuring effects interference in a quantum system. The results offer insight into the non-classical nature of optical lights and quantum measurement. It also equips us the knowledge to create two qubit quantum states with single photons through spatial and polarization entanglement.

3 Procedure

This experiment was conducted into Mach-Zehnder Interferometer (MZI) configured to explore the effects of spatial-polarization entanglement and quantum interference. A continuous laser with a power output of ~15 mW is aligned to generate single photons via spontaneous down conversion (SPDC). Coincidence counts between signal and idler photons will be used to measure quantum interference.

The input polarization of the photon is fixed to a vertical state using a linear polarizer. The spatial and polarization degrees of freedom are controlled using half wave plates, phase delays, and a detection polarizer.

First, the MZI is aligned with both HWPs at 0 degrees, and the detector polarizer set to transmit vertically polarized photons. Background and accidental coincidence counts were measured with the pump laser off to account for detector noise. Second, we will measure beeline interference. With the HWP's at 0 degrees, the system will be configured such that path information is not introduced. The piezoelectric phase shifters will scan the relative phase δ between the interferometer arms, and coincidence counts will be recorded to establish baseline interference. Then, we will conduct controlled path labeling. Here, the HWP of the bottom arm of the MZI will be incrementally rotated to 15, 30, and 45 degrees. This will introduce polarization path information to our existing system. For each angle, the phase will be measured, and coincidence counts will be recorded to see the changing interference. Finally, the half wave plate will be set to 45 degrees. The detection polarizer will be rotated to 45 degrees, producing an output state into the diagonal polarization basis. The phase will be measured once again. Coincidence counts will be measured to see if interference is restored by erasing path information.

All measurements will be recorded and accidental/background data should be subtracted from the raw data collected to obtain true quantum measurements. Data is plotted in graphs correlating phase and coincidence counts. Finally, visibility should be tracked through different measurements to quantify the presence of quantum interference.

4 Results and Discussion

4.1 Measurement 1

To start, measurements were taken with the MZI half-wave plate set to 0°, meaning that the MZI was operating normally. Background/accidental counts were measured to be 3167 for the signal counts and 0 for the coincidence counts. After sweeping through two periods of the MZI, Figure 1 was obtained. Just like in the previous lab, we observe interference fringes as we step the phase difference between the two MZI arms.

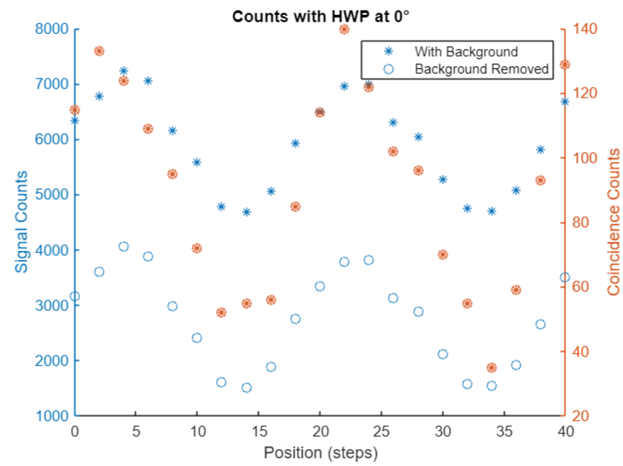

Figure 1. Counts with HWP at 0°

The visibility is calculated as:

$$V = \frac{4067-1523}{4067+1523} = 0.455$$

This is not far off of what it should theoretically be for perfect interference, which is 0.5. It may be slightly off due to reflections, absorption by polarizers, or slight misalignment of polarization elements.

4.2 Measurement 2

The previous measurement was then repeated with the HWP in one of the MZI's arms rotated to 15°, resulting in a 30° shift of the light's polarization. We observe that the max measured signal counts decrease from 4067 to 3354. This is expected because, as we change the polarization of one arm's light, it begins interfering with the other arm less and less of it gets through the polarizer before the detector. We also observe the visibility decrease to 0.436, which matches our expectation for the visibility to go down as the amount of interference decreases.

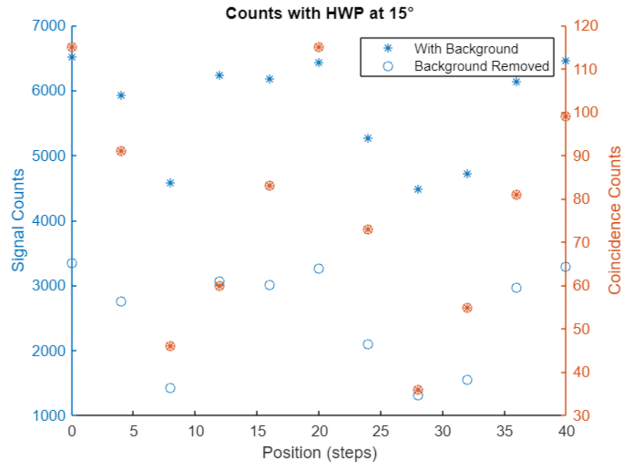

Figure 2. Counts with HWP at 15°

4.3 Measurement 3

Once again, we rotated the HWP another 15°, resulting in the light's polarization in that arm now being rotated 60°. We observe a decrease in max signal counts to 2213 and visibility to 0.402, which is once again expected as described in Section 4.2.

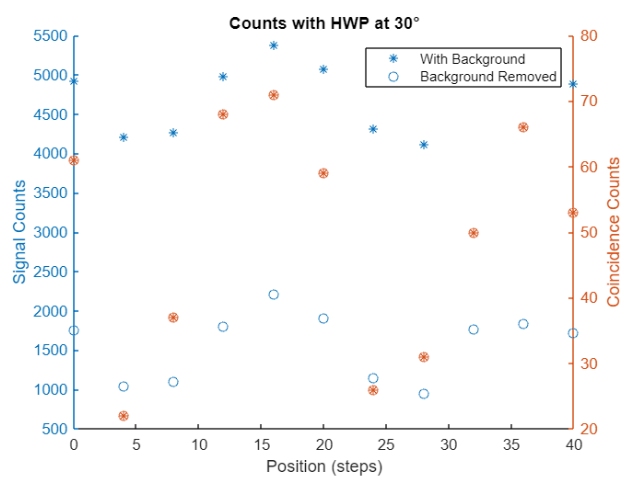

Figure 3. Counts with HWP at 30°

4.4 Measurement 4

Finally, we rotated the HWP to 45°, resulting in the light's polarization in that arm now being rotated 90°. We observe a decrease in max signal counts to 1238 and visibility to 0.216 as expected. It should be noted that the minimum value can reasonably be considered an outlier due to noise, as it is substantially lower than the other measurements in this section, as seen in Figure 4. Without this data point, the visibility ends up being 0.0545.

An important result of this measurement is that there is no visible interference pattern formed anymore. This result is expected because interference no longer occurs between two waves, or wavefunctions, if their polarizations differ by 90°. This results in our paths becoming distinguishable from one another.

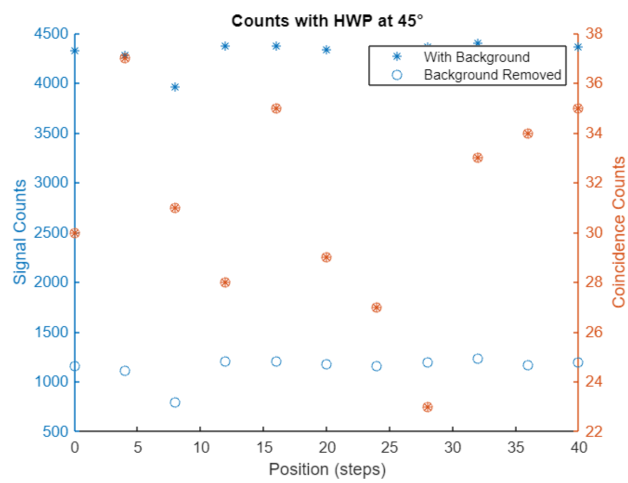

Figure 4. Counts with HWP at 45°

4.5 Measurement 5

The previous experiment, with the HWP at 45°, was redone with the detector's linear polarizer set to 45°. Unlike the previous experiment, we see the interference pattern reappear after rotating the linear polarizer, causing the paths to no longer be distinguishable. This result is called the "Quantum Eraser," as we have erased the path information by rotating the polarizer. This result happens due to the linear polarizer at the detector causing the wavefunctions to occupy the same polarization at the measurement (detector) rather than being at two different polarizations with 90° spacing as seen in the previous experiment.

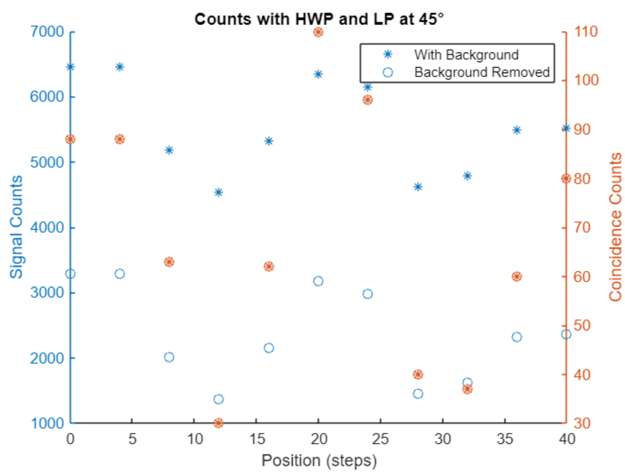

Figure 5. Counts with HWP and LP at 45°

The eigenstate of the final state for this experimental setup is calculated as:

$$\hat{E_{xy}}\hat{Z}'\left|\psi, D\right\rangle = \frac{1}{2\sqrt{2}}\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} i\left(e^{i\delta}+1\right) \\ i\left(1-e^{i\delta}\right) \\ e^{i\delta}-1 \\ -1-e^{i\delta} \end{pmatrix}$$

$$\hat{E_{xy}}\hat{Z}'\left|\psi, D\right\rangle = \frac{1}{2\sqrt{2}}\begin{pmatrix} \sqrt{2}i \\ \sqrt{2}i \\ e^{i\delta}-1 \\ -1-e^{i\delta} \end{pmatrix}$$

5 Conclusion

This experiment successfully demonstrated the quantum eraser effect using a Mach-Zehnder Interferometer. By controlling the polarization state of photons through HWP's and analyzing the interference visibility, we observed how having knowledge of which path the photon travelled through, affected our coincidence counts. When path information was introduced through the change in angle within the MZI HWP, interference patterns became less noticeable. Conversely, by changing the basis at which we measure the output of the MZI, "which path" information could be effectively erased, to restore interference, highlighting the capabilities of linear optics in quantum computing.

These results not only reinforce key principles of quantum mechanics but also demonstrate how quantum states can be engineered and manipulated, allowing for applications in quantum computing.